

Evolutionary Algorithms for Actuator-Sensor-Communication Co-Design in Distributed Control

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Abstract—This paper studies the co-design of actuators, sensors, and communication in the distributed setting, where a networked plant is partitioned into subsystems each equipped with a sub-controller interacting with other sub-controllers. The objective is to jointly minimize control cost (measured by LQ cost) and material cost (measured by the number of actuators, sensors, and communication links used). We approach this using an evolutionary algorithm to selectively prune a baseline dense LQR controller. We provide convergence bounds, performance bounds, and probabilistic stability guarantees for this algorithm. For unstable plants, controller pruning is more likely to induce instability; we provide an algorithm modification to address this. The proposed method is validated in simulations. One key result is that co-design of a 98-state swing equation model can be done on a standard laptop in seconds; the co-design outperforms naive controller pruning by X%.

Index Terms—component, formatting, style, styling, insert

I. INTRODUCTION

Large-scale networked dynamical systems arise in a wide range of modern engineering applications, including power grids, traffic networks, multi-agent robotic systems, and cyber-physical infrastructures. In such systems, centralized control architectures are often impractical due to scalability limitations, communication constraints, and robustness concerns. These challenges have motivated extensive research on *distributed control*, where control actions are computed using only local information and limited inter-agent communication. In parallel, practical deployments increasingly demand *co-design* methodologies that jointly consider control performance, communication structure, sensing and actuation placement, and implementation cost, rather than treating these aspects in isolation.

Despite significant progress, *distributed controller co-design* remains a fundamental and largely unresolved problem. Classical distributed control frameworks typically assume that the information or communication structure is fixed a priori and focus on synthesizing an optimal controller within that prescribed structure [1]. More recent approaches, such as System Level Synthesis (SLS) [2], provide scalable and convex formulations for distributed controller design under locality constraints, but still rely on a pre-specified sparsity parameterization. In contrast, co-design problems pose a more challenging question: *which feedback links, sensing channels, or actuation couplings should exist in the first place, and how should they be selected to balance performance and sparsity?* This question inherently couples discrete combinatorial

decisions with continuous controller optimization, leading to highly nonconvex and combinatorial problem formulations.

Several lines of related work have attempted to address aspects of this challenge. Sparsity-promoting optimal control methods introduce regularization or convex relaxations to encourage sparse feedback gains [1], enabling partial structural selection [3] while retaining tractable optimization. Other approaches study communication-aware or delay-aware controller synthesis, often focusing on specific network models or restrictive assumptions. While these methods have yielded valuable insights, they typically decouple structure selection from controller synthesis, rely on convex surrogates that may be conservative, or face scalability limitations in large-scale networks. As a result, a general and flexible framework for distributed control co-design that simultaneously explores structural and control dimensions remains elusive.

Distributed control co-design problems, in which control synthesis is coupled with the selection of sensing, actuation, or communication structures, have received increasing attention in recent years. A representative line of work studies the *simultaneous sensor and actuator selection problem* (SSASP), where the objective is to identify a minimal set of sensing and actuation nodes that renders a large-scale dynamical network stabilizable. In [4], the authors formulate SSASP under static output feedback as a mixed-integer nonconvex optimization problem and develop exact and heuristic algorithms based on linear matrix inequalities and combinatorial search. While these methods provide valuable insights into joint structure-controller design, they primarily focus on stabilization feasibility and rely on solving a large number of LMIs, which doesn't take into account the controller performance optimization and can limit scalability in high-dimensional systems.

Complementary to this algorithmic perspective, structural and graph-theoretic approaches have been proposed to study control and communication co-design at an abstract level. For instance, [4] investigates the joint selection of actuators, sensors, and control configurations using structural system theory, and establishes fundamental complexity limits for such co-design problems.

Overall, existing approaches either rely on exact or approximate combinatorial optimization with significant computational overhead, or focus on structural feasibility without explicitly optimizing closed-loop feedback controller performance. In this work, we explore an alternative perspective based on *evolutionary algorithms* (EA) for distributed control co-design. Rather than enforcing a fixed sparsity pattern or relying on convex relaxations, we treat the co-design problem

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as a population-based dynamic search over both controller parameters and structural attributes. Evolutionary algorithms are well suited for this setting as they naturally handle mixed discrete–continuous decision variables, admit flexible objective formulations, and can incorporate stability and performance constraints through selection mechanisms.

II. PROBLEM SETUP: SPARSE LQR CO-DESIGN VIA EVOLUTIONARY ALGORITHM

A. System Model

Consider a discrete-time linear time-invariant system:

$$x_{k+1} = Ax_k + Bu_k \quad (1)$$

where $x_k \in \mathbb{R}^{N_x}$ is the state vector, $u_k \in \mathbb{R}^{N_u}$ is the control input, $A \in \mathbb{R}^{N_x \times N_x}$ is the state matrix, and $B \in \mathbb{R}^{N_x \times N_u}$ is the input matrix. We seek a sparse state-feedback controller:

$$u_k = K_{\text{sparse}} x_k \quad (2)$$

where $K_{\text{sparse}} \in \mathbb{R}^{N_u \times N_x}$ is constructed to performance metrics, such as the LQR cost.

B. Design Variables

The gene vector is:

$$\theta = [n_{\text{links}}, a_1, \dots, a_{N_u}, s_1, \dots, s_{N_x}]^\top, \quad (3)$$

with $n_{\text{links}} \in \{n_{\min}, \dots, n_{\max}\}$, and $a_j, s_i \in \{0, 1\}$.

C. Objective Function

Here, we take the example of the LQR cost objective and Q, R are both identity matrix. For a stabilizing controller K , the infinite-horizon LQR cost is

$$J_{\text{LQR}}(K) = \sum_{k=0}^{\infty} (x_k^\top x_k + u_k^\top u_k), \quad u_k = Kx_k, \quad (4)$$

which corresponds to the identity weights $Q = I, R = I$. The benchmark cost J_{BM} corresponds to the optimal dense LQR controller

$$K_{\text{dense}} = -\text{dlqr}(A, B, I, I),$$

i.e.,

$$J_{\text{BM}} = J_{\text{LQR}}(K_{\text{dense}}).$$

Define $\mathcal{I}_m$ as the index set of the $m = n_{\text{links}}$ largest-magnitude entries of K_{dense} . Then

$$K_{\text{sparse}}(\theta) = \Pi_{a,s}(\mathcal{P}_{\mathcal{I}_m}(K_{\text{dense}})),$$

where $\mathcal{P}_{\mathcal{I}_m}$ keeps entries in $\mathcal{I}_m$ and zeros out the rest, and $\Pi_{a,s}$ applies the actuator/sensor masks.

$$J_{\text{total}}(\theta) = \frac{J_{\text{LQR}}(K_{\text{sparse}}(\theta))}{J_{\text{BM}}} + (w_c \|K_{\text{sparse}}\|_0 + w_a \|a\|_1 + w_s \|s\|_1). \quad (5)$$

example 1 (4 states, 2 actuators). Consider the dense LQR gain

$$K_{\text{dense}} = \begin{bmatrix} 0.8 & -0.5 & 0.1 & 0.05 \\ -0.6 & 0.4 & -0.2 & 0.03 \end{bmatrix}.$$

Step 1: Top- m sparsification. Let $m = 3$. The three largest-magnitude entries are

$$(1, 1) = 0.8, \quad (2, 1) = -0.6, \quad (1, 2) = -0.5.$$

Keeping only these entries yields

$$K_{\text{top-3}} = \begin{bmatrix} 0.8 & -0.5 & 0 & 0 \\ -0.6 & 0 & 0 & 0 \end{bmatrix}.$$

Step 2: Apply actuator and sensor masks. Let the actuator mask be

$$a = [1, 0]^\top,$$

and the sensor mask be

$$s = [1, 1, 0, 1]^\top.$$

After row/column projection,

$$K_{\text{sparse}} = \begin{bmatrix} 0.8 & -0.5 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Hence, K_{sparse} is obtained by magnitude-based truncation followed by structural masking.

D. Optimization Problem

$$\begin{aligned} \min_{\theta} \quad & J_{\text{total}}(\theta) \\ \text{s.t.} \quad & \rho(A + BK_{\text{sparse}}) < 1 \\ & n_{\text{links}} \in \{n_{\min}, \dots, n_{\max}\}, \quad a_j, s_i \in \{0, 1\} \end{aligned} \quad (6)$$

where $\rho(\cdot)$ denotes the spectral radius. The problem is solved using an evolutionary algorithm introduced in the next section.

III. EVOLUTIONARY ALGORITHM FOR SPARSE LQR CO-DESIGN

As shown in Algorithm ??, given system matrices (A, B, Q, R) and a benchmark dense LQR cost J_{BM} which is drawn from optimal LQR solution K^* , the population is initialized with randomly generated genes. At each generation, every individual is decoded into a sparse constraint for the feedback controller and evaluated via its closed-loop cost J_i . Individuals yielding unstable closed-loop dynamics are penalized.

The population of binary mask vectors is ranked by ascending fitness cost. The top n_{top} individuals—those achieving the lowest combined LQR performance and sparsity penalty—are carried over unchanged into the next generation as elites, ensuring the best-discovered controller structures are never lost. The remaining population slots are filled by offspring produced through single-point crossover and bit-flip mutation. Two parent chromosomes are selected uniformly at random from the current population; with probability p_{cross} , a crossover point is drawn uniformly along the binary gene vector, and the child inherits the prefix from one parent and the suffix from the other. Each gene component—corresponding to a single element of the gain matrix K within k -hop graph distance—is then independently flipped with probability p_{mut} ,

Algorithm 1 EA-Based Sparse LQR Co-Design (Refined)

Require: A, B, Q, R , EA params $(N_p, G_{\max}, p_c, p_m, n_e)$, trade-off α

Ensure: K^*, θ^*

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1: Initialize:
2:  $K_d \leftarrow \text{dlqr}(A, B, Q, R)$ ,
    $J_{bm} \leftarrow \text{cost\_LQR}(A, B, Q, R, K_{bm})$ ;
3: Generate  $\mathcal{P}_0 = \{\theta_i\}_{i=1}^{N_p}$  with  $\theta_i = [\ell_i, a_i, s_i]$ ;
4: for  $t = 0$  to  $G_{\max} - 1$  do
5:   for each  $\theta_i \in \mathcal{P}_t$  do
6:      $K_{s,i} \leftarrow \Pi_{a_i, s_i}(\text{trunc}(K_d, \ell_i))$ ;
7:     if  $\rho(A + BK_{s,i}) \geq 1$  then
8:        $F_i = 10^9$ ; ▷ Stability penalty
9:     else
10:       $F_i \leftarrow \text{cost\_EA}(K_{s,i}, J_{bm}, \alpha)$ ;
11:    end if
12:  end for
13:   $\mathcal{P}_{\text{elite}} \leftarrow \text{select } n_e \text{ best individuals from } \mathcal{P}_t$ ;
14:  for  $j = n_e + 1$  to  $N_p$  do
15:     $\theta_{p1}, \theta_{p2} \leftarrow \text{Selection}(\mathcal{P}_t)$ ;
16:     $\theta_{\text{new}} \leftarrow \text{Mutation}(\text{Crossover}(\theta_{p1}, \theta_{p2}, p_c), p_m)$ ;
17:     $\mathcal{P}_{t+1} \leftarrow \text{Project}(\theta_{\text{new}}, \mathcal{B})$ ;
18:  end for
19: end for
20: return  $\theta^* = \arg \min F(\theta)$ ,  $K^* = \text{Decode}(\theta^*)$ .
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toggling that element between retained (copying the benchmark LQR gain value) and zeroed out. Gene bits beyond the k -hop neighborhood remain fixed at zero throughout evolution.

Throughout evolution, the minimum cost, average cost, and ratio of unstable individuals are recorded. After $G_{\max}$ generations, the best-performing individual is returned, yielding a near-optimal sparse LQR controller that balances performance and structural complexity.

IV. EA ALGORITHM FOR STABLE PLANTS

In this section, we will prove the stability of the near optimal solution generated by the EA. First, we will provide some reasonable assumptions for the system we consider. Then, we will use the lemma from [5] to attain the bound on the dense optimal LQR result. Finally, we will prove that our EA algorithm maintains stability with an exponentially increasing probability.

Definition 1 (Stability, Stabilizability, and Detectability). *Let $L > 0$ and $\alpha \in [0, 1)$. We define the following:*

- (a) Φ is (L, α) -**stable** if $\|\Phi^t\| \leq L\alpha^t$ for all $t \in \mathbb{I}_{\geq 0}$.
- (b) (A, B) is (L, α) -**stabilizable** if there exists a feedback gain $\bar{K}$ such that $\|\bar{K}\| \leq L$ and $A - B\bar{K}$ is (L, α) -stable.
- (c) (A, C) is (L, α) -**detectable** if $(A^\top, C^\top)$ is (L, α) -stabilizable.

Assumption 1. *There exist constants $L > 1$, $\alpha \in (0, 1)$, and $\gamma \in (0, 1)$ such that:*

- (a) $\|A\|, \|B\|, \|Q\|, \|R\| \leq L$;

- (b) $R \succeq \gamma I$;
- (c) (A, B) is (L, α) -stabilizable;
- (d) $Q \succeq 0$, and $(A, Q^{1/2})$ is (L, α) -detectable.

Lemma 1 (Exponential Decay of Optimal Gain). *Under Assumption 1, the optimal gain K^* satisfies $\|K_{ij}^*\| \leq \Upsilon \rho^{d_g(i,j)}$, where:*

$$\gamma_F := \frac{(1-\alpha)^2}{L^2(1+L)^2}, \quad \gamma_G := \frac{(1-\alpha)^2\gamma}{2L^4(1+L)^2}, \quad L_P := \frac{L^3(1+L^2)}{1-\alpha^2}, \quad (3.1a)$$

$$\bar{\mu} := (2L_H^2/\gamma_G + \gamma_G + L_H)/\gamma_F, \quad (3.1b)$$

$$\gamma_H := \left(\frac{2}{\gamma_G} + \left(1 + \frac{4L_H}{\gamma_G} + \frac{4L_H^2}{\gamma_G^2} \right) \frac{L_H(1 + \bar{\mu}L_H)}{\gamma_F} + \bar{\mu} \right)^{-1}, \quad (3.1c)$$

$$L_H := \max(2L + 1, L_P + 1), \quad (3.1d)$$

$$\rho := \left(\frac{L_H^2 - \gamma_H^2}{L_H^2 + \gamma_H^2} \right)^{1/2}, \quad \Upsilon := \frac{L_H}{\gamma_H^2 \rho}.$$

Furthermore, $\rho \in (0, 1)$ and $\Upsilon \geq 1$.

The proof is in [5]. Here, $d_g(i, j)$ denotes the graph distance (or shortest path distance) between nodes i and j in the communication graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, which is defined as the minimum number of edges connecting them.

Assumption 2. *The graph growth is polynomially bounded. Specifically, for any node $i \in \mathcal{V}$, the number of nodes at distance d satisfies:*

$$|\{j \in \mathcal{V} : d_g(i, j) = d\}| \leq d^2, \quad \forall d \in \mathbb{I}_{\geq 0}. \quad (7)$$

A. Probabilistic Stability Guarantee

1) *Setting and Notation:* Consider the discrete-time plant

$$x_{k+1} = Ax_k + Bu_k, \quad u_k = Kx_k, \quad (8)$$

with closed-loop matrix $A_K := A + BK$. We call K (Schur) stabilizing if $\rho(A_K) < 1$ and define the (spectral-radius) stability slack

$$\mu(K) := 1 - \rho(A_K) > 0. \quad (9)$$

For a diagonalizable reference $A_K = V\Lambda V^{-1}$, denote $\kappa(V) := \|V\|_2 \|V^{-1}\|_2$.

2) *EA Offspring: Crossover + Structure-Aware Sparsification:* Each gene encodes diagonal weights (Q, R) (positive definite) and a sparsity operator that produces a sparse controller from the dense LQR gain $K_{\text{dense}}(Q, R)$. In the EA, an offspring is formed by (i) near-zero mutation and (ii) crossover on a_i, s_j .

mask-based truncation (e.g., intersection mask / top- m selection) yielding

$$K_{\text{sp}} = K_{\text{dense}} - \Delta K, \quad \tau := \frac{\|\Delta K\|_F}{\|K_{\text{dense}}\|_F}. \quad (10)$$

Here we make two assumptions to set guaranties of stability as well as the norm bound on ΔK when updating the binary mask vector.

Assumption 3 (Local continuity of LQR map). *Fix a stabilizing parent (Q_i, R_i) producing a stable sparse closed loop $A_i := A + BK_{i,\text{sp}}$ with margin $\mu_i := 1 - \rho(A_i) > 0$. Near-zero mutation implies $\delta_{K,i}$ is small.*

The discarded part ΔK has approximately isotropic orientation relative to dominant eigenmodes of A_i , so that for any $\delta \in (0, 1)$,

$$\|\Delta K\|_2 \leq c \frac{\|\Delta K\|_F}{\sqrt{d}} \sqrt{\log(1/\delta)} \quad \text{with probability } \geq 1 - \delta, \quad d := \min(N_u, N_x). \quad (11)$$

for some moderate constant c .

Theorem 1 (High-probability stability of EA offspring). *Let parent i be chosen (e.g., the one with larger margin μ_i). Consider an offspring constructed by near-zero-mutation crossover (Q', R') and sparsification $K_{\text{sp}} = K_{\text{dense}}(Q', R') - \Delta K$. Assume A_i is diagonalizable with condition number $\kappa(V_i)$ and define the total perturbation on the closed loop by*

$$A_{\text{sp}} = A_i + B\Delta K_{\text{total}} \quad (12)$$

$$\Delta K_{\text{total}} := (K_{\text{dense}}(Q', R') - K_{\text{dense}}(Q_i, R_i)) + \Delta K. \quad (13)$$

Then, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\rho(A_{\text{sp}}) \leq \rho(A_i) + \kappa(V_i)\|B\|_2 \left(\delta_{K,i} + c \frac{\|\Delta K\|_F}{\sqrt{d}} \sqrt{\log(1/\delta)} \right). \quad (14)$$

Consequently, a sufficient condition for $\rho(A_{\text{sp}}) < 1$ with probability $\geq 1 - \delta$ is

$$\kappa(V_i)\|B\|_2 \left(\delta_{K,i} + c \frac{\|\Delta K\|_F}{\sqrt{d}} \sqrt{\log(1/\delta)} \right) < \mu_i. \quad (15)$$

Equivalently, writing $\|\Delta K\|_F = \tau \|K_{\text{dense}}\|_F$, one obtains the explicit (tail-ratio) success probability bound

$$\mathbb{P}[\rho(A_{\text{sp}}) < 1] \geq 1 - \exp \left(- \left[\frac{(\mu_i - \kappa(V_i)\|B\|_2 \delta_{K,i}) \sqrt{d}}{c \kappa(V_i)\|B\|_2 \|K_{\text{dense}}\|_F \tau} \right]^2 \right), \quad (16)$$

whenever $\mu_i > \kappa(V_i)\|B\|_2 \delta_{K,i}$.

Corollary 1 (Weighted-error implies probabilistic stability surrogate). *If $\|K_{\text{sp}} - K_{\text{dense}}\|_F$ is controlled by the weighted proxy (e.g., via design choice of $\{\alpha_\ell\}$ and bandlimit w), then the tail-ratio bound in Theorem 1 yields a monotone relationship*

$$\varepsilon_\alpha \downarrow \implies \mathbb{P}[\rho(A + BK_{\text{sp}}) < 1] \uparrow. \quad (17)$$

In particular, as an empirically calibrated robust surrogate (Sec. 6),

$$\mathbb{P}[\rho(A + BK_{\text{sp}}) < 1 \mid \varepsilon_\alpha] \gtrsim 1 - \beta \varepsilon_\alpha, \quad (18)$$

where $\beta > 0$ is system-dependent and can be estimated from Monte-Carlo trials or pseudospectral sensitivity.

V. GERSHGORIN-BASED STABILITY REPAIR

a) *Motivation:* In the EA pipeline, each individual's gene encodes weighting matrices (Q, R) and a sparsity pattern. A dense stabilizing controller $K_{\text{ref}} = -\text{dlqr}(A, B, Q, R)$ is first computed, then *truncated* to ℓ nonzero entries and masked by actuator/sensor binary vectors. This sparsification can destroy closed-loop stability: $\rho(A + BK_{\text{ref}}) < 1$ does not imply $\rho(A + BK_{\text{sp}}) < 1$. Without repair, unstable individuals receive a large penalty and are effectively wasted, slowing convergence. We therefore seek a *repair operator* that adjusts the *values* of the nonzero entries of K_{sp} (without changing which entries are nonzero) to recover stability, while staying as close as possible to the original truncated controller.

b) *Standing Assumptions:* Now we make some mild assumptions for the gershgorin repair method. Let $n_x = \dim(x)$, $n_u = \dim(u)$, and define $A_{\text{cl}}(K) \triangleq A + BK$.

Assumption 4 (Stabilizability). *The pair (A, B) is stabilizable, i.e., there exists K such that $\rho(A_{\text{cl}}(K)) < 1$.*

Assumption 5 (Reference Controller). *There exists a dense stabilizing centralized LQR optimal controller K_{ref} satisfying $\rho(A_{\text{cl}}(K_{\text{ref}})) < 1$.*

Assumption 6 (Fixed Sparsity Pattern). *Let*

$$\mathcal{S} = \{(u, j) : K_{\text{sp}}(u, j) \neq 0\}$$

denote the admissible support. Any admissible output controller K_{out} must satisfy

$$\text{supp}(K_{\text{out}}) \subseteq \mathcal{S}.$$

Let $n_x = \dim(x)$, $n_u = \dim(u)$, and write $A_{\text{cl}}(K) \triangleq A + BK$.

c) *Notation:* For a matrix $M \in \mathbb{R}^{n \times n}$, define the Gershgorin row-sum

$$R_i(M) \triangleq \sum_{j=1}^n |M_{ij}|, \quad i = 1, \dots, n, \quad (19)$$

and the Gershgorin radius $\bar{R}(M) \triangleq \max_i R_i(M)$. For a gain matrix $K \in \mathbb{R}^{n_u \times n_x}$, write $\bar{R}(K) \triangleq \bar{R}(A + BK)$ when A, B are clear from context. Denote the sparsity-constrained set $\mathcal{K}_{\mathcal{S}} \triangleq \{K \in \mathbb{R}^{n_u \times n_x} : \text{supp}(K) \subseteq \mathcal{S}\}$.

Lemma 2 (Gershgorin sufficient condition). *If $\bar{R}(K) < 1$, then $\rho(A + BK) < 1$, i.e., the closed-loop system is Schur stable.*

Proof. By the Gershgorin disk theorem, every eigenvalue λ of $A_{\text{cl}} = A + BK$ lies in at least one disk $\mathcal{D}_i = \{z \in \mathbb{C} : |z - A_{\text{cl},ii}| \leq r_i\}$ where $r_i = \sum_{j \neq i} |A_{\text{cl},ij}|$. Hence $|\lambda| \leq |A_{\text{cl},ii}| + r_i = R_i(A_{\text{cl}}) \leq \bar{R}(K) < 1$. $\square$

Proposition 1 (Convexity). *For each i , the function $K \mapsto R_i(A + BK)$ is convex. Consequently, the Gershgorin-stable set $\mathcal{G} \triangleq \{K : \bar{R}(K) < 1\}$ is a convex (open) subset of $\mathbb{R}^{n_u \times n_x}$, and $\mathcal{G} \cap \mathcal{K}_{\mathcal{S}}$ is convex.*

Proof. Fix row i . For each column j , $A_{\text{cl},ij} = A_{ij} + \sum_{u=1}^{n_u} B_{iu} K_{uj}$ is affine in K . Therefore $|A_{\text{cl},ij}|$ is convex in

K (absolute value of an affine function). $R_i = \sum_j |A_{\text{cl},ij}|$ is a non-negative sum of convex functions, hence convex. $\bar{R} = \max_i R_i$ is the pointwise maximum of convex functions, hence convex. The sublevel set $\{K : \bar{R}(K) < 1\}$ is therefore convex. Intersecting with the affine subspace $\mathcal{K}_S$ preserves convexity. $\square$

d) *Subgradient of $\bar{R}(K)$:*

Proposition 2. Let $i^* = \arg \max_i R_i(A + BK)$ and $A_{\text{cl}} = A + BK$. A subgradient of $\bar{R}$ with respect to K_{uj} is

$$g_{uj} = \text{sign}(A_{\text{cl},i^*j}) \cdot B_{i^*u}. \quad (20)$$

Under the sparsity constraint 4, the projected subgradient is $\tilde{g}_{uj} = g_{uj} \mathbf{1}_{(u,j) \in \mathcal{S}}$.

Proof. Since $\bar{R}(K) = \max_i R_i(K)$, a subgradient of $\bar{R}$ at K can be taken as any subgradient of R_{i^*} at K [?, Sec. 3.1.2]. Now,

$$R_{i^*}(K) = \sum_{j=1}^{n_x} |A_{i^*j} + (BK)_{i^*j}| = \sum_{j=1}^{n_x} |A_{i^*j} + \sum_u B_{i^*u} K_{uj}|.$$

Each summand $\phi_j(K) = |A_{i^*j} + \sum_u B_{i^*u} K_{uj}|$ is the absolute value of an affine function $\ell_j(K)$. When $\ell_j(K) \neq 0$, ϕ_j is differentiable with $\partial \phi_j / \partial K_{uj} = \text{sign}(\ell_j) \cdot B_{i^*u}$. When $\ell_j(K) = 0$, any value in $[-|B_{i^*u}|, |B_{i^*u}|]$ is a valid subgradient element. Taking $\text{sign}(0) = 0$ (as in the implementation) yields a valid choice. Summing over j gives the subgradient of R_{i^*} ; however, since each K_{uj} appears only in the j -th summand, the subgradient with respect to K_{uj} reduces to (20). Projecting onto $\mathcal{K}_S$ simply zeros out components outside the support. $\square$

e) *Repair algorithm.*: Given K_{sp} with $\rho(A + BK_{\text{sp}}) \geq 1$ and K_{ref} with $\rho(A + BK_{\text{ref}}) < 1$, the repair operator proceeds in three phases of decreasing sparsity fidelity.

Phase 1 (Subgradient projection, sparsity-preserving). Solve the convex feasibility problem

$$\text{find } K \in \mathcal{K}_S \quad \text{s.t.} \quad \bar{R}(K) \leq \rho^*, \quad (21)$$

where $\rho^* < 1$ is a target (e.g. 0.95), via the iteration

$$K^{(t+1)} = \Pi_S \left[K^{(t)} - \eta_t \tilde{g}^{(t)} \right], \quad (22)$$

with Polyak step size

$$\eta_t = \frac{\bar{R}(K^{(t)}) - \rho^*}{\|\tilde{g}^{(t)}\|_F^2}, \quad (23)$$

and Π_S the projection onto $\mathcal{K}_S$ (i.e., zeroing out entries outside $\mathcal{S}$).

Proposition 3 (Phase 1 convergence). If $\mathcal{G} \cap \mathcal{K}_S \neq \emptyset$ (i.e., a Gershgorin-stable controller exists on the given sparsity pattern), then the Polyak subgradient iteration (22)–(23) generates a sequence $\{K^{(t)}\}$ satisfying

$$\min_{0 \leq \tau \leq t} \bar{R}(K^{(\tau)}) - \rho^* \leq \frac{\|K^{(0)} - K^*\|_F^2}{2 \sum_{\tau=0}^t \eta_\tau} \xrightarrow{t \rightarrow \infty} 0, \quad (24)$$

where $K^* \in \mathcal{G} \cap \mathcal{K}_S$ is any feasible point. In particular, the iterates reach $\bar{R} < 1$ in finitely many steps, which by Lemma 2 guarantees Schur stability.

Proof. This is a standard result for Polyak-step subgradient methods applied to convex feasibility [?]. Let K^* be any point with $\bar{R}(K^*) \leq \rho^*$. By convexity of $\bar{R}$:

$$\bar{R}(K^{(t)}) - \rho^* \leq \bar{R}(K^{(t)}) - \bar{R}(K^*) \leq \langle \tilde{g}^{(t)}, K^{(t)} - K^* \rangle_F.$$

From the update rule and the non-expansiveness of Π_S :

$$\begin{aligned} \|K^{(t+1)} - K^*\|_F^2 &\leq \|K^{(t)} - \eta_t \tilde{g}^{(t)} - K^*\|_F^2 \\ &= \|K^{(t)} - K^*\|_F^2 - 2\eta_t \langle \tilde{g}^{(t)}, K^{(t)} - K^* \rangle_F \\ &\quad + \eta_t^2 \|\tilde{g}^{(t)}\|_F^2. \end{aligned}$$

Substituting the Polyak step size $\eta_t = (\bar{R}(K^{(t)}) - \rho^*) / \|\tilde{g}^{(t)}\|_F^2$ and using the subgradient inequality:

$$\|K^{(t+1)} - K^*\|_F^2 \leq \|K^{(t)} - K^*\|_F^2 - \frac{(\bar{R}(K^{(t)}) - \rho^*)^2}{\|\tilde{g}^{(t)}\|_F^2}.$$

Therefore $\{\|K^{(t)} - K^*\|_F^2\}$ is nonincreasing and

$$\sum_{t=0}^{\infty} \frac{(\bar{R}(K^{(t)}) - \rho^*)^2}{\|\tilde{g}^{(t)}\|_F^2} \leq \|K^{(0)} - K^*\|_F^2 < \infty.$$

Since $\|\tilde{g}^{(t)}\|_F$ is bounded (by $\|B\|_F \sqrt{n_x}$), the numerator $(\bar{R}(K^{(t)}) - \rho^*)^2 \rightarrow 0$, establishing convergence. The rate (24) follows from the standard telescoping argument for subgradient methods. $\square$

Phase 2 (Line search on support, sparsity-preserving). If Phase 1 reduces $\bar{R}$ but does not certify $\rho(A + BK) < 1$ within the iteration budget (recall the Gershgorin condition is sufficient but not necessary), we exploit the reference controller. Define $K_{\text{ref}}^S \triangleq \Pi_S(K_{\text{ref}})$ and search

$$K(\eta) = (1 - \eta) K^{(\text{best})} + \eta K_{\text{ref}}^S, \quad \eta \in [0, 1], \quad (25)$$

via bisection for the smallest η with $\rho(A + BK(\eta)) < 1$. Since the support is fixed at $\mathcal{S}$ throughout, the sparsity pattern is exactly preserved.

Phase 3 (Line search + re-sparsification, fallback). If Phase 2 fails (the projection K_{ref}^S may itself be unstable), we interpolate toward the full K_{ref} :

$$K(\eta) = (1 - \eta) K^{(\text{best})} + \eta K_{\text{ref}}, \quad (26)$$

then re-sparsify $K(\eta)$ by retaining its ℓ largest-magnitude entries (matching the original link budget). Since K_{ref} is Schur-stabilizing and the set of stabilizing gains is open, there exists $\eta^* > 0$ such that $K(\eta)$ remains stable for all η sufficiently close to 1. Bisection finds the smallest such η to maximize structural fidelity.

Proposition 4 (Phase 3 existence guarantee). Under 4–5, Phase 3 always finds a stabilizing K_{out} for sufficiently fine bisection resolution.

Proof. At $\eta = 1$, $K(1) = K_{\text{ref}}$ is stabilizing by 5, so $\rho(A+BK(1)) < 1$. Since eigenvalues depend continuously on matrix entries, there exists $\delta > 0$ such that $\rho(A+BK_{\text{sp}}) < 1$ for any K_{sp} with $\|K_{\text{sp}} - K_{\text{ref}}\| < \delta$. As $\eta \rightarrow 1$, the re-sparsified matrix converges to the top- ℓ truncation of K_{ref} , whose perturbation from K_{ref} is bounded. Hence the bisection terminates successfully for η sufficiently close to 1. $\square$

f) Complexity. Each subgradient iteration costs $\mathcal{O}(n_u n_x)$ for the outer-product gradient and masking; the periodic eigenvalue check costs $\mathcal{O}(n_x^3)$ but is performed only every 10 iterations. Phases 2–3 each require at most 25 eigenvalue computations ($25 \times \mathcal{O}(n_x^3)$). The total per-individual repair cost is therefore $\mathcal{O}(T_{\text{max}} n_u n_x + n_x^3)$, which is dominated by the `dlqr` call already present in the EA evaluation.

VI. CONVERGENCE OF EA UNDER LIPSCHITZ SENSITIVITY AND TOPOLOGY DECAY

There are multiple ways to show the global convergence/complexity of the gradient descent method for the mere LQR problem [6]. We consider the discrete-time LTI system above. The evolutionary algorithm's objective is the normalized performance term plus structural penalties. The algorithm evolves diagonal weights (Q, R) , a link budget ℓ , and actuator/sensor masks, then forms a sparse controller by taking the dense LQR gain $K_{\text{dense}} = -\text{dlqr}(A, B, Q, R)$ and applying a top- ℓ truncation and masks; denote the resulting composite map by $K = K(x)$ for a gene x .

a) Assumption 1 (Lipschitz sensitivity on $\mathcal{S}$): Assume there exists a stable subset $\mathcal{S}$ with uniform margin $\rho(A+BK) \leq 1 - \delta$ for all $K \in \mathcal{S}$.

A. Convergence rate bound analysis

On $\mathcal{S}$, the LQR cost has Lipschitz gradient with respect to the sparse vector $k = \text{vec}(K)$:

$$\|\nabla_k J(K_1) - \nabla_k J(K_2)\| \leq L_J \|K_1 - K_2\|_F, \quad \forall K_1, K_2 \in \mathcal{S}. \quad (27)$$

In particular, if $K^* \in \mathcal{S}$ is a first-order stationary point of the stabilizing LQR objective (so $\nabla_k J(K^*) = 0$), then for any $\hat{K} \in \mathcal{S}$,

$$J(\hat{K}) - J(K^*) \leq \frac{L_J}{2} \|\hat{K} - K^*\|_F^2. \quad (28)$$

We recall the following result from [?].

a) lemma 2 (topology and exponential decay): The system topology is known with hop distance $d(i, j)$ and the (reference) optimal gain satisfies the proved decay bound

$$|K^*(i, j)| \leq C e^{-d(i, j)}. \quad (29)$$

Let K_h^* denote the hop-truncated controller defined by

$$K_h^*(i, j) = K^*(i, j) \mathbf{1}\{d(i, j) \leq h\}.$$

Then the truncation error admits the bound

$$\|K^* - K_h^*\|_F^2 = \sum_{d(i, j) > h} |K^*(i, j)|^2 \leq C^2 \sum_{r=h+1}^{\infty} N_{\Delta}(r) e^{-2r}, \quad (30)$$

where $N_{\Delta}(r) := \#\{(i, j) : d(i, j) = r\}$ depends only on the growth of the underlying graph. For bounded-degree graphs (including lattices and grids), $N_{\Delta}(r)$ grows at most exponentially in r , and thus the above tail sum decays exponentially in h . Consequently, there exists a constant $\tilde{C} > 0$ such that

$$\|K^* - K_h^*\|_F \leq \tilde{C} e^{-h}. \quad (31)$$

Combining this estimate with the Lipschitz gradient property in Assumption 1 yields the following performance-gap bound for the hop-truncated controller (assuming $K_h^* \in \mathcal{S}$):

$$J(K_h^*) - J(K^*) \leq \frac{L_J}{2} \tilde{C}^2 e^{-2h}. \quad (32)$$

b) Convergence rate lower bound.: Under Assumptions A1–A2, define the optimality gap $\Delta(t) \triangleq J_{\min}(t) - J_{\mathcal{R}}^*$ with $J_{\mathcal{R}}^*$ the best cost reachable by recombination alone. Elitism guarantees $\Delta(t) \geq 0$ is nonincreasing. At generation t , let $q(t)$ denote the probability that at least one of the $P - n_{\text{Top}}$ crossover offspring improves the current best. Under A1, an offspring whose mask differs from the current best by Hamming distance δ can reduce cost by at most $L\delta$, so that the per-offspring improvement probability satisfies $p_{\text{imp}}(t) \leq c \Delta(t)$ for a constant c depending on L , d_m , and current allele frequencies. Consequently $q(t) \leq 1 - (1 - c \Delta(t))^{P - n_{\text{Top}}}$, and the gap satisfies

$$\mathbb{E}[\Delta(t+1)] \geq (1 - q(t)) \Delta(t) \geq \Delta(0) \prod_{\tau=0}^{t-1} (1 - q(\tau)). \quad (33)$$

Under A2, diversity decays monotonically, so $q(t) \rightarrow 0$ and one obtains the exponential lower envelope

$$J_{\min}(t) \geq J_{\mathcal{R}}^* + A_0 e^{-\bar{\lambda} t}, \quad \bar{\lambda} \leq (P - n_{\text{Top}}) p_c L \bar{d}_0, \quad (34)$$

where $\bar{d}_0$ is the initial mean Hamming diversity and p_c the crossover rate.

Empirical exponential fits on a 4×4 grid ($d_m = 48$, $P = 20$, $n_{\text{Top}} = 10$) yield $\lambda \in [0.035, 0.069]$ across eight seeds, consistent with (34): the observed rate never exceeds the theoretical ceiling $\bar{\lambda}$, confirming that zero-mutation convergence speed is fundamentally limited by the initial population diversity.

B. Numerical Experiment on convergence rate

We evaluate the convergence behavior and structural robustness of the proposed EA-LQR co-design framework on a 4×4 grid system.

To Multi-Seed Robustness.

Figure ?? summarizes convergence results across eight independent random seeds. The best-cost trajectories exhibit consistent monotonic decrease and converge to a narrow band of final values. The mean $\pm$ std envelope further confirms low inter-seed variability after approximately 80–100 generations. While transient instability events occur during early exploration, the number of unstable individuals per generation remains bounded and does not prevent convergence. The final-cost boxplot indicates small dispersion across seeds, demonstrating robustness of the evolutionary search against stochastic initialization.

VII. PERFORMANCE BOUNDS RELATIVE TO BASELINE

We compare three controller design strategies evaluated under the composite cost (5) with $\alpha = 0$:

- 1) **Dense LQR:** $K = \text{dlqr}(A, B, I, I)$ with entries $|K_{ij}| < 10^{-3}$ zeroed out. This retains near-optimal LQR performance ($J_{\text{perf}} \approx 1$) but incurs high structural cost due to $\text{nnz}(K) \approx mn$.
- 2) **Diagonal (local) LQR:** Each actuator i feeds back only from its own node states (x_{2i-1}, x_{2i}) . Structural cost is minimal ($\text{nnz}(K) = 2m$) but the absence of inter-node coordination degrades J_{perf} .
- 3) **EA-LQR:** The proposed co-design with Gershgorin stability repair (Algorithm 2).

Proposition 5 (Dense LQR upper bound). *For the composite cost (5) with $\alpha = 0$ and Dense LQR controller K_d (with entries $|K_{ij}| \geq 10^{-3}$),*

$$J(K_d) \leq 1 + w_c mn + w_r m + w_s n = 1 + 0.05 mn + 0.4 m + 0.2 n, \quad (35)$$

where n is the state dimension and m is the input dimension.

Proof. Since K_d is obtained by thresholding the LQR-optimal K^* at 10^{-3} , we have $J_{\text{perf}}(K_d) \approx \|P_{K_d}\|_F^2 / \|P^*\|_F^2 \leq 1 + \epsilon$ for negligible ϵ . The structural cost is bounded by the worst case $\text{nnz}(K_d) \leq mn$, with all m actuators and n sensors active. Summing yields (35). $\square$

Proposition 6 (Diagonal LQR decomposition). *For the diagonal (local) controller K_{diag} where actuator i uses only states (x_{2i-1}, x_{2i}) ,*

$$J(K_{\text{diag}}) = J_{\text{perf}}^{\text{diag}} + 0.9 m, \quad (36)$$

where $J_{\text{perf}}^{\text{diag}} \geq 1$ with equality only when the off-diagonal entries of K^* are identically zero (i.e., the system is already decoupled). For coupled networks, $J_{\text{perf}}^{\text{diag}}$ grows with the algebraic connectivity $\lambda_2(\mathcal{L})$.

Proof. The structural cost is exactly $w_c \cdot 2m + w_r \cdot m + w_s \cdot 2m = (0.1 + 0.4 + 0.4)m = 0.9m$. The performance cost satisfies $J_{\text{perf}}^{\text{diag}} = \|P_{K_{\text{diag}}}\|_F^2 / \|P^*\|_F^2$. By the perturbation bound for discrete Lyapunov equations, $\|P_{K_{\text{diag}}} - P^*\| \geq c \|\Delta K\|$ where $\Delta K = K_{\text{diag}} - K^*$. For coupled networks ΔK contains all off-diagonal inter-node feedback entries of K^* , whose magnitude grows with the graph Laplacian connectivity. $\square$

Proposition 7 (EA dominance). *Let K_{EA}^* denote the EA output with sparsity $\ell = \text{nnz}(K_{\text{EA}}^*)$, actuator set $\mathcal{A}$, sensor set $\mathcal{S}$. Then*

$$J(K_{\text{EA}}^*) \leq \min\{J(K_d), J(K_{\text{diag}})\}. \quad (37)$$

Moreover, EA improves over Dense LQR whenever the structural savings exceed the performance degradation:

$$\underbrace{J_{\text{perf}}^{\text{EA}} - 1}_{\Delta_{\text{perf}}} < \underbrace{w_c(\text{nnz}(K_d) - \ell) + w_r(m - |\mathcal{A}|) + w_s(n - |\mathcal{S}|)}_{\Delta_{\text{struct}}}. \quad (38)$$

Proof. Both K_d and K_{diag} are feasible for the EA (representable by appropriate gene settings), so elitism guarantees $J(K_{\text{EA}}^*) \leq \min\{J(K_d), J(K_{\text{diag}})\}$. Inequality (38) follows

TABLE I
NORMALIZED COST J/J_{dense} AND STRUCTURAL STATISTICS (MEAN OVER 3 SEEDS, $\alpha = 0$).

Grid	Dense LQR		Diagonal LQR		EA-LQR	
	J/J_d	nnz	J/J_d	nnz	J/J_d	nnz
3×3	1.000	90	0.728	18	0.526	7
4×4	1.000	192	0.666	32	0.389	13
5×5	1.000	235	0.721	50	0.386	17
7×7	1.000	747	0.585	98	0.282	39

by expanding $J(K_{\text{EA}}^*) < J(K_d)$ into performance and structural components. $\square$

Theorem 2 (Automatic Decentralization). *If the system is weakly coupled such that the performance gain from any inter-node link $e = (i, j)$ ($i \neq j$) is bounded by the communication cost, i.e., $|\Delta J_{\text{perf}}(e)| < w_c$, then the optimal EA controller reduces to a distributed structure:*

$$\text{supp}(K_{\text{EA}}^*) \subseteq \mathcal{S}_{\text{diag}}. \quad (39)$$

Furthermore, EA implicitly optimizes the topology by potentially pruning redundant local feedback loops (diagonal elements) that do not contribute significantly to stability or performance, yielding a topology strictly sparser than standard Diagonal LQR.

Table I and Fig. 1–2 summarize the results across grids $\{3 \times 3, 4 \times 4, 5 \times 5, 7 \times 7\}$. Three observations emerge:

(i) **EA always improves over both baselines.** Across all grid sizes, EA achieves $J/J_{\text{dense}} < 1$, reducing cost by 47–72% over dense LQR and 28–52% over diagonal LQR (Fig. 1). Dense LQR suffers from excessive structural penalties ($J_{\text{struct}} \gg J_{\text{perf}}$), while diagonal LQR pays a heavy LQR performance penalty due to the loss of inter-node feedback. EA balances both components through joint optimization of the sparsity pattern, actuator mask, and sensor mask.

(ii) **The advantage grows with system dimension.** As n increases, the dense controller’s structural cost scales as $O(mn)$ while diagonal cost stays $O(m)$. EA exploits the expanding combinatorial space to find increasingly efficient sparse topologies, widening the gap to both baselines (Fig. 2a–b).

(iii) **Convergence is fast relative to the search space.** EA surpasses the dense baseline within ~ 1 generations on average (Fig. 2c), despite the sparsity pattern space growing combinatorially. The Gershgorin repair mechanism ensures that most individuals are evaluated with finite cost, providing strong selection signal even in early generations.

VIII. SIMULATIONS

We did the experiment on Gershgorin repair algorithm 2 for 3 random seed systems which are unstable open loop (spectral radius is scaled to 1.08) and have 25 nodes. The result is as follows 3:

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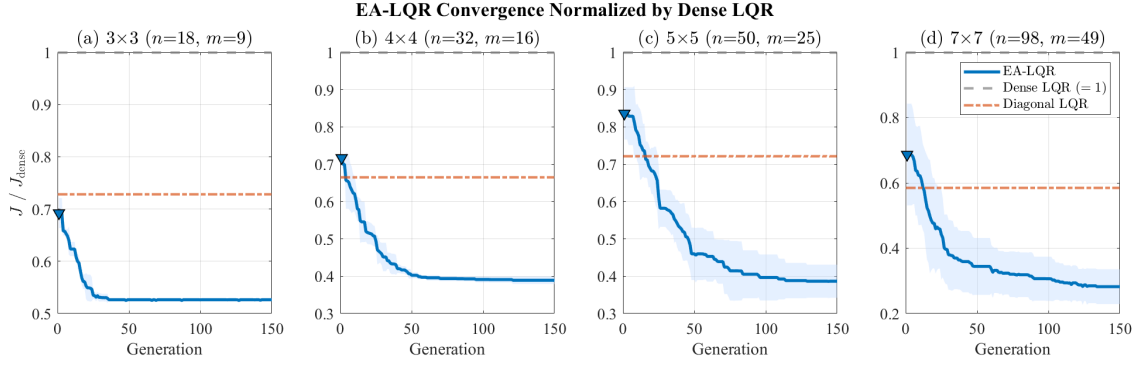


Fig. 1. Normalized EA convergence (J/J_{dense}) across grid sizes. Gray dashed: dense LQR (= 1); orange dash-dotted: diagonal LQR. Triangle: generation at which EA first beats dense LQR. Shaded: $\pm 1\sigma$ over 3 seeds.

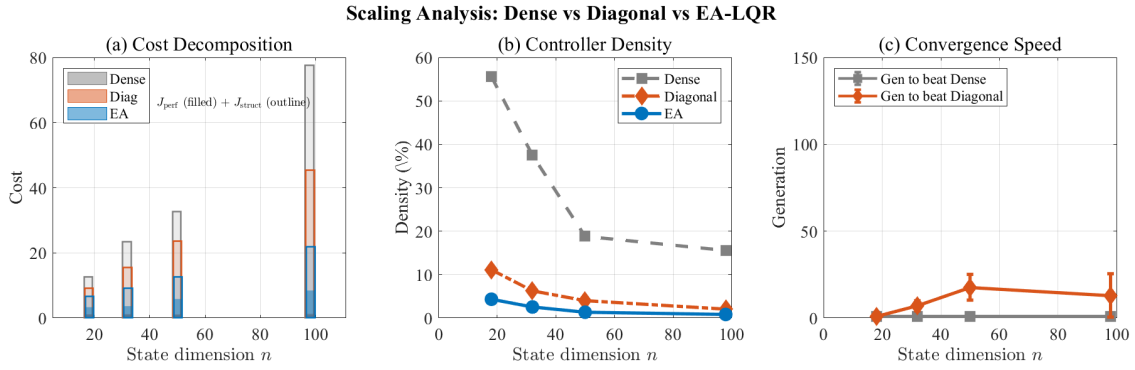


Fig. 2. Scaling analysis. (a) Cost decomposition: filled bars are J_{perf} , outlines show total J . (b) Controller density ($\text{nnz}(K)/mn$). (c) Generations for EA to surpass each baseline.

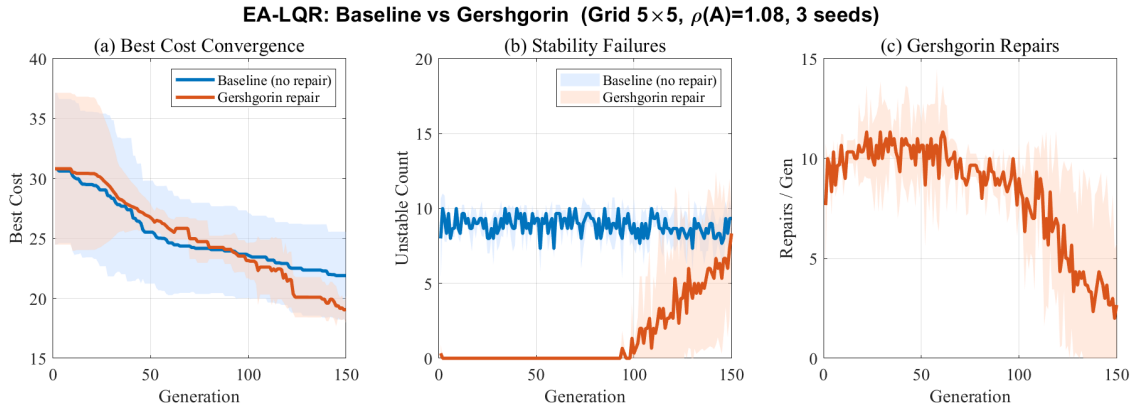


Fig. 3. Average cost, unstable individual controllers and successfully repaired controllers.

Algorithm 2 Gershgorin Three-Phase Stability Repair

Require: State matrix $A \in \mathbb{R}^{n \times n}$, input matrix $B \in \mathbb{R}^{n \times m}$, sparse controller $K \in \mathbb{R}^{m \times n}$ with $\rho(A + BK) \geq 1$, reference stabilizing controller K^{ref} (e.g., from LQR), target row-sum $\bar{\rho} < 1$, max iterations T

Ensure: Stabilized controller K^{out} with $\rho(A + BK^{\text{out}}) < 1$
1: $\mathcal{S} \leftarrow \{(u, j) : K_{uj} \neq 0\}$ $\triangleright$ sparsity support
2: $\ell \leftarrow |\mathcal{S}|$ $\triangleright$ link count

— **Phase 1: Subgradient descent on Gershgorin row-sum** —

3: **for** $t = 1, \dots, T$ **do**
4: $A_{\text{cl}} \leftarrow A + BK$
5: $R_i \leftarrow \sum_{j=1}^n |[A_{\text{cl}}]_{ij}|$, $i = 1, \dots, n$ $\triangleright$ Gershgorin row-sums
6: $i^* \leftarrow \arg \max_i R_i$
7: **if** $R_{i^*} < \bar{\rho}$ **then break** $\triangleright$ converged
8: **end if**
9: $g_{uj} \leftarrow \text{sign}([A_{\text{cl}}]_{i^*j}) \cdot B_{i^*u}$, $(u, j) \in \mathcal{S}$ $\triangleright$ subgradient
10: $\eta \leftarrow \min\left(\frac{R_{i^*} - \bar{\rho}}{\|g\|_F^2}, 0.5\right)$ $\triangleright$ Polyak step with clamp
11: $K \leftarrow K - \eta g$
12: **end for**
13: $K^{(1)} \leftarrow \arg \min_{K' \in \{K_t\}} \rho(A + BK')$ $\triangleright$ best iterate
14: **if** $\rho(A + BK^{(1)}) < 1$ **then**
15: **return** $K^{\text{out}} \leftarrow K^{(1)}$
16: **end if**

— **Phase 2: Line search on sparsity support toward K^{ref}** —

17: $K_{\mathcal{S}}^{\text{ref}} \leftarrow K^{\text{ref}} \odot \mathbf{1}_{\mathcal{S}}$ $\triangleright$ project K^{ref} onto $\mathcal{S}$
18: Binary search over $\alpha \in [0, 1]$:
19: $K(\alpha) = (1 - \alpha) K^{(1)} + \alpha K_{\mathcal{S}}^{\text{ref}}$
20: Find minimal α^* such that $\rho(A + BK(\alpha^*)) < 1$
21: **if** found **then**
22: **return** $K^{\text{out}} \leftarrow K(\alpha^*)$
23: **end if**

— **Phase 3: Line search toward full K^{ref} + re-sparsification** —

24: Binary search over $\alpha \in [0, 1]$:
25: $\tilde{K}(\alpha) = (1 - \alpha) K^{(1)} + \alpha K^{\text{ref}}$
26: $K_{\text{sp}}(\alpha) \leftarrow \text{Top}_{\ell}(\tilde{K}(\alpha))$ $\triangleright$ keep ℓ largest-magnitude entries
27: Find minimal α^* such that $\rho(A + B K_{\text{sp}}(\alpha^*)) < 1$
28: **if** found **then**
29: **return** $K^{\text{out}} \leftarrow K_{\text{sp}}(\alpha^*)$
30: **end if**

31: **return** $K^{\text{out}} \leftarrow K$ $\triangleright$ all phases failed
